

Detecting Multilingual, Multicultural And Multievent Online Polarization

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Abstract

As online discourse becomes increasingly fractured, the automated detection of ideological polarization is critical for understanding political divide and improving content moderation. This study utilizes the English subset of the POLAR benchmark to address two key tasks: binary polarization detection and multi-label classification. We fine-tuned large pre-trained language models, namely BERT-Large-Uncased and RoBERTa-Large, and applied class-weighted and focal loss strategies to address class imbalance. Our results show that fine-tuned LLMs are effective for polarization detection, with RoBERTa-Large achieving a macro F1-score of 0.8548, while multi-label classification remains more challenging, reaching 0.4643 macro F1. Class-weighting improves performance on minority categories despite a 15:1 imbalance. While the results demonstrate the effectiveness of fine-tuned transformer models, rare and overlapping polarization categories remain a challenge for future work.

1 Introduction

Online social media platforms have become a central space for the public to express opinions. However, underlying user biases combined with lack of filter while sharing thoughts online, mostly without repercussions, results in content that is frequently polarized and extreme. Such discourse can lead to stereotyping, intolerance, and broader societal conflict. Detecting polarization in online text is challenging due to the frequent use of implicit language, sarcasm, and informal writing styles.

In this work, we study the task of polarization detection and classification using the POLAR (Detecting Multilingual, Multicultural and Multievent Online Polarization) benchmark (Naseem et al., 2025), focusing on its English subset. The task is divided into two subtasks. Subtask 1 addresses binary classification, determining whether a text is polarized

or not. Subtask 2 formulates a multi-label classification problem, covering political, racial/ethnic, religious, gender/sexual orientation, and other forms of polarization.

A key challenge of the POLAR dataset is class imbalance. While political polarization is highly dominant, other categories are substantially under-represented. In addition, many polarized instances show label co-occurrence. These factors make it difficult to achieve balanced performance, particularly when using macro-averaged evaluation metrics that penalize poor performance on minority classes.

This project aims to answer the research question *How effectively can large language models combined with NLP principles detect and categorize different types of polarization in English messages from online forums?* To address this, we fine-tune large pre-trained transformer-based language models, specifically BERT-Large-Uncased (Devlin et al., 2019a) and RoBERTa-Large (Liu et al., 2019), using task-specific classification heads. Our approach provides an evaluation of these models for both binary and multi-label polarization detection. We further explore strategies to mitigate class imbalance and improve performance on rare polarization types while pointing out remaining challenges such as implicit polarization, unclear labels, and possible annotator subjectivity for future work.

2 Related Work

Previous works on hate speech detection, particularly in multimodal contexts, typically highlight that models trained only on text usually have difficulties capturing contextual and nuanced signals of harmful or polarized content. For example, Boishakhi et al. (Boishakhi et al., 2023) demonstrate that single-modality systems (e.g., text-only) tend to underperform compared to approaches that include additional prompts such as facial expres-

sions or vocal tone. Their work shows that the integration of multiple modalities lead to a more robust detection, even if classical machine learning methods are used. This reinforces the idea that linguistic signals alone are not sufficiently reliable on their own to capture complex social phenomena such as hate or hostility towards someone.

This limitation is further emphasized in other studies on text-based hate speech classification using traditional and deep learning models. Warner and Hirschberg (Warner and Hirschberg, 2012), as well as Badjatiya et al. (Badjatiya et al., 2017), describe that, although high accuracy is achievable on well-balanced datasets, model performance drops significantly in the presence of label imbalance, domain shifts, or nuanced categories of abuse. In other racism and sexism detection tasks, some similar observations appear (Kshirsagar et al., 2018), here, models show strong performance on common categories, but reduced results on less common or context-dependent cases.

To address these difficulties mentioned, recent approaches suggest Transformer architectures. BERT, introduced by Devlin et al. (Devlin et al., 2019b), revolutionized the field by using a masked language modeling objective that allows the model to learn deep bidirectional representations of the text. BERT can capture complicated semantic relationships that unidirectional models are not able to catch, this is possible due to it conditioning context across all layers on the left and right. RoBERTa builds from this foundation and introduces some key refinements. Liu et al. (Liu et al., 2019) showed that, by removing the Next Sentence Prediction task, training on much larger datasets (160 Gb vs 16 Gb for BERT), and implementing a dynamic mask which changes every epoch, the model achieves a better understanding of the language. Due to this refinements, RoBERTa is effective for identification of subtle signals of polarization in social media texts.

3 Dataset

This project utilizes the English subset of the POLAR (Detecting Multilingual, Multicultural and Multievent Online Polarization) benchmark (Naseem et al., 2025). To align with the research goals outlined in the Introduction, we conducted a quantitative analysis focusing on the structural challenges of the data, specifically class imbalance and label co-occurrence.

3.1 Data Source and Overview

The data are sourced from the POLAR benchmark, which aims to identify online polarization across diverse platforms and events. We utilized the official training split provided by the benchmark (15% for test). In this study, we focus on the English subset to perform binary polarization detection (Subtask 1) and multi-label polarization type classification (Subtask 2).

The training dataset consists of 2,670 instances collected from various online sources, including Reddit, Bluesky, and news forums. These instances cover real-world events such as global elections and regional conflicts, reflecting the "multi-event" nature of the task. The average text length is 12.38 words, often characterized by the informal writing styles and implicit language mentioned in Section 1.

3.2 Subtask 1: Polarization Detection

Subtask 1 involves binary classification to determine whether a text is *Polarized* (1) or *Not Polarized* (0). The distribution of the 2,670 training instances is as follows:

- **Not Polarized (0):** 1,674 instances (62.7%)
- **Polarized (1):** 996 instances (37.3%)

This inherent class imbalance represents the first challenge, as models may naturally lean toward the majority class without specific optimization strategies.

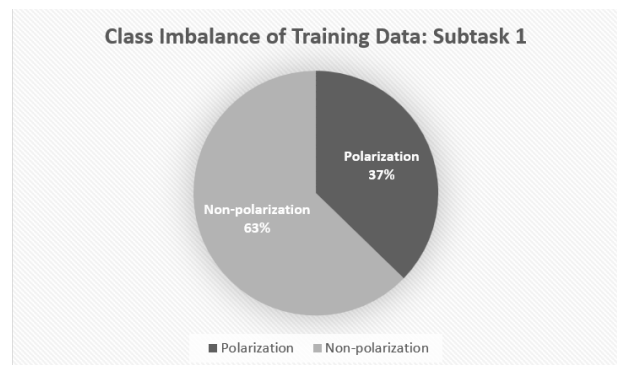


Figure 1: Distribution of Polarized vs. Not Polarized instances in the English training set.

3.3 Subtask 2: Polarization Type Classification (Multi-label)

For instances identified as polarized, Subtask 2 requires predicting one or more of five defined types:

Political, Racial/Ethnic, Religious, Gender/Sexual, and Other. Our exploration identified two primary hurdles in this subtask:

Extreme Class Imbalance: The dataset suffers from severe label sparsity. While the Political category is highly dominant (989 instances), the Gender/Sexual category is substantially underrepresented with only 65 instances, resulting in an imbalance ratio of approximately 15:1. This imbalance may bias the model toward the majority categories, potentially inflating overall accuracy while masking poor performance on minority polarization types. These observations motivated our choice of macro-F1 as the primary evaluation metric and the use of class-weighting and Focal Loss strategies during training.

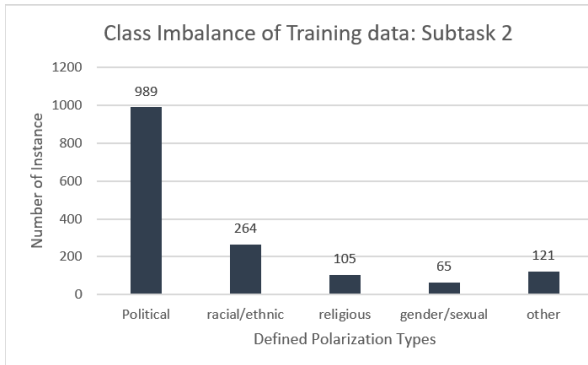


Figure 2: Distribution of 5 polarization categories of the instances.

Label Co-occurrence: Polarization in online discourse is rarely isolated. Analysis shows that **40.96%** (408 instances) of the polarized data are annotated with multiple labels (see Figure ??). This "intersectional" property demonstrates that many messages span multiple social dimensions, confirming the necessity of a multi-label classification framework that can capture overlapping thematic signals.

4 Methodology

4.1 Model Architecture

Our architecture utilizes a transfer learning framework based on the Transformer encoder-decoder paradigm. We implement a modular pipeline consisting of a pre-trained backbone followed by a task-specific classification head as visualized in Figure 4

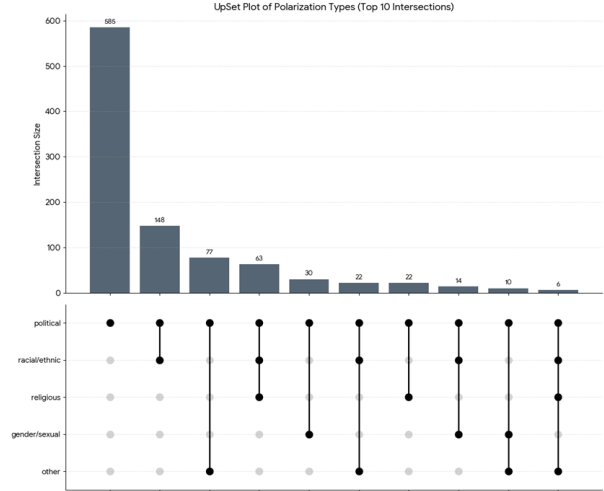


Figure 3: Label co-occurrence of the polarized data

4.2 Input Representation and Tokenization

Raw text sequences are processed using model-specific tokenizers (WordPiece for BERT and Byte-Level BPE for RoBERTa). We utilize *dynamic padding* via a specialized data collator, which pads sequences to the maximum length of the current batch rather than a global fixed length. This optimizes memory throughput and preserves the positional integrity of the $[CLS]$ token, which serves as the aggregate sequence representation.

4.3 Transformer Backbone

We evaluate two primary backbones:

- **BERT-Large-Uncased:** A 24-layer encoder with 1024 hidden units and 16 attention heads. This model leverages a lowercased vocabulary, prioritizing semantic content over stylistic casing.
- **RoBERTa-Large:** A robustly optimized variant of BERT that utilizes dynamic masking and a larger byte-level vocabulary, providing more nuanced embeddings for detecting subtle ideological shifts.

4.4 Classification Head and Loss Optimization

A task-specific classification head is attached to the final hidden representation of the $[CLS]$ token. For Subtask 1, the classifier produces a single logit; for Subtask 2, it outputs a 5-dimensional logit vector.

Given an input text x , the model produces logits z , and probabilities are computed via the sigmoid

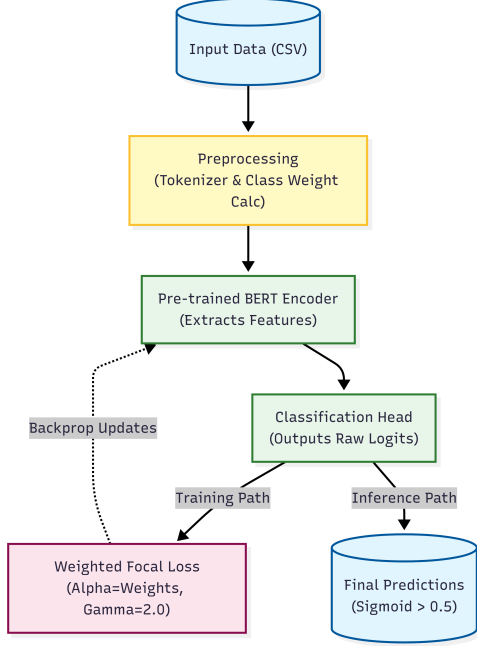


Figure 4: Architectural pipeline for online polarization detection. The workflow shows the transition from raw text input through model-specific tokenization and Transformer encoding (BERT/RoBERTa), ending in a task-specific classification head that utilizes weighted loss functions to address class imbalance

function $p = \sigma(z) = 1/(1 + e^{-z})$. Since multi-label predictions are independent, we apply sigmoid to each label instead of softmax, and produce final predictions by thresholding at 0.5.

4.5 Handling Class Imbalance

The POLAR dataset exhibits severe imbalance, particularly in Subtask 2, as described in Section 3. To mitigate this effect, we adopt class-weighted optimization and Focal Loss.

4.5.1 Class-Weighted Loss

For multi-label classification we compute a weight for each class $w_c = \frac{N}{n_c}$, where N denotes the total number of training samples and n_c is the number of positive instances for class c . The weights are normalized by their mean to stabilize optimization and are incorporated through the `pos_weight` parameter of `BCEWithLogitsLoss`.

4.5.2 Focal Loss

We further apply Focal Loss (Lin et al., 2018) to improve learning on minority labels. The objective is defined as

$$\mathcal{L}_{\text{focal}} = (1 - p_t)^\gamma \mathcal{L}_{\text{BCE}},$$

where

$$p_t = \begin{cases} p, & \text{if } y = 1, \\ 1 - p, & \text{if } y = 0, \end{cases}$$

and $\gamma = 2.0$ is the focusing parameter. This formulation down-weights easy examples and encourages the model to focus on hard or rare samples.

4.6 Training Procedure

We fine-tune each model end-to-end using the HuggingFace Trainer API. For each epoch the model is evaluated on a validation set, and the best model is automatically restored.

Early stopping is used with a patience of 2 epochs based on validation macro-F1.

4.7 Evaluation Metric

The primary evaluation metric is the macro-averaged F1-score, $F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c$, which equally weights all classes and is therefore suitable for imbalanced data, as it penalizes poor performance on minority categories.

For Subtask 2, predictions are obtained by applying a sigmoid activation followed by independent thresholding at 0.5 for each label.

5 Experimental Setup

5.1 Data Partitioning

To ensure strong model evaluation and hyperparameter tuning, we partition the official training set into distinct training and validation subsets using a 15% split ratio. A fixed random seed of 42 is maintained across all experiments to ensure reproducibility.

For **Subtask 1** (Binary Detection), we employ a *stratified split* to maintain the original distribution of polarized versus non-polarized samples across both partitions, mitigating the risk of class representational bias. Conversely, for **Subtask 2** (Multi-label Classification), we utilize a *standard random split*. Due to the multi-label nature of the task, where the overlapping combinations of several categories make traditional single-label stratification mathematically non-trivial.

5.2 Hyperparameters

Training is performed with the hyperparameters (Loshchilov and Hutter, 2018), given by the following Table 1:

The smaller batch size in Subtask 2 is necessary due to increased memory usage from multi-label outputs and class weighting.

Table 1: Hyperparameter Configuration and Task-Specific Batch Sizes

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	2×10^{-5}
Maximum Epochs	7
Early Stopping Patience	2 Epochs
Max Sequence Length	128 Tokens
Task	Batch Size
Subtask 1 (Binary)	16
Subtask 2 (Multi-label)	8

5.3 Prediction and Thresholding

Model outputs are converted to probabilities using a sigmoid function. Predictions are obtained using a fixed threshold:

$$\hat{y}_c = \begin{cases} 1 & \text{if } p_c \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

5.4 Model Selection

At each epoch the validation macro-F1 is computed. The model checkpoint with the highest validation macro-F1 is automatically selected and saved, ensuring fair comparison between BERT-Large and RoBERTa-Large.

6 Analysis and Discussion

To evaluate model convergence and the impact of our class-balancing strategies, we employ a multi-faceted diagnostic pipeline. Training and validation dynamics are monitored via **loss and macro-F1 curves** to detect overfitting and ensure stable fine-tuning. Given the significant class imbalance in the POLAR dataset, we further analyze performance through **confusion matrices**. For the binary detection task, a standard matrix is used to identify false positive trends, while for the multi-label task, we generate **per-class confusion matrices** to isolate the model’s sensitivity to rare polarization categories. Table ?? summarizes the performance of the fine-tuned models. The binary task is solved reliably by large transformer models, while the multi-label task remains significantly more difficult. RoBERTa-Large achieves a macro F1 of 0.8548 on polarization detection, confirming that contextual language models can effectively recognize polarized tone even in short informal messages. However, performance drops substantially in Subtask 2, where the macro F1 decreases to 0.4643,

indicating that identifying the *type* of polarization is a much harder semantic problem than detecting its presence.

Table 2: Performance comparison of evaluated models on the POLAR English subset

Model	Binary Macro F1	Multi-label Macro F1
RoBERTa-Large	0.8548	0.4643
BERT-Large-Uncased	0.8129	0.3358
DeBERTa-v3-base	0.8257	0.3479

6.1 Training Behavior

The learning curves in Figure 5, 6, 7, 8 show a

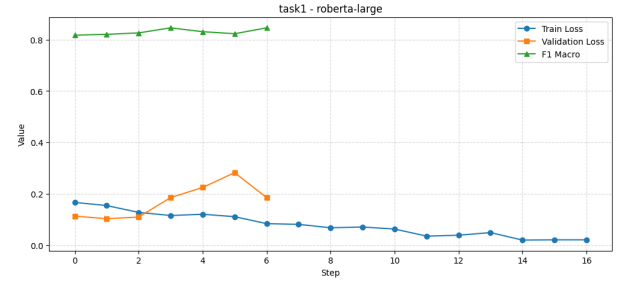


Figure 5: Performance of RoBERTa-Large for POLAR subtask-1

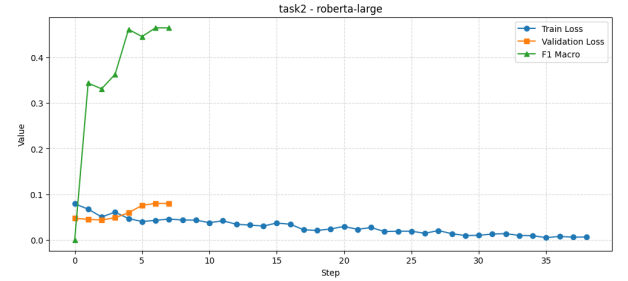


Figure 6: Performance of RoBERTa-Large for POLAR subtask-2

stable fine-tuning process. Training loss decreases rapidly within the first few epochs, while validation macro-F1 plateaus early. This pattern is typical for transformer fine-tuning on small datasets. The models quickly learn general sentiment and stance cues but struggle to further improve on subtle semantic distinctions. Early stopping prevents clear overfitting; after a few epochs, validation performance stabilizes even though training loss continues to decrease.

RoBERTa consistently converges to a better validation score than BERT. This difference is expected because RoBERTa is trained with dynamic masking and larger training corpora, which improves contextual sensitivity. In short texts averaging only about

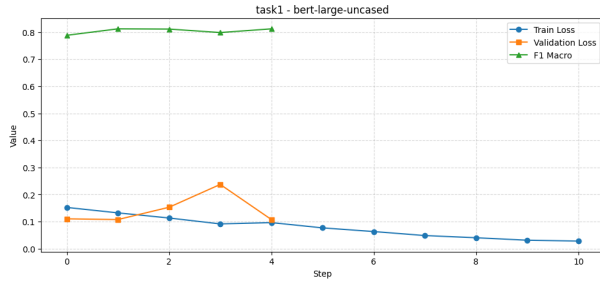


Figure 7: Performance of BERTa-Large for POLAR subtask-1

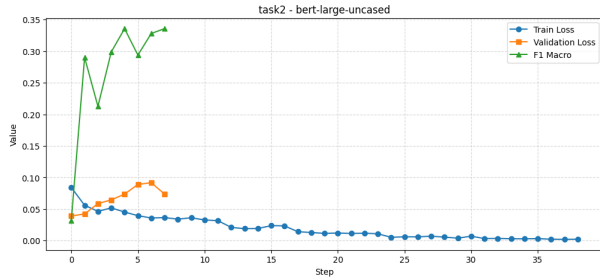


Figure 8: Performance of BERT-Large for POLAR subtask-2

12 words, every token carries high importance, and better contextual representations translate directly into improved classification performance.

The DeBERTa results show competitive behavior but slightly worse final performance than RoBERTa. Although DeBERTa’s disentangled attention is theoretically advantageous, the relatively small dataset appears insufficient for it to fully exploit this architectural strength.

6.2 Binary Task: Confusion Matrix Analysis

The confusion matrix for Subtask 1 (Figure 12) reveals an important property of the classifier: the model is conservative in predicting polarization. The majority of errors are false negatives rather than false positives. In other words, the model often predicts *not polarized* when the text is actually polarized. This behavior is directly connected to both dataset structure and language characteristics. Polarized messages in the dataset frequently rely on implication rather than explicit hostility. For example, sarcastic remarks, coded language, and indirect references do not always contain obvious sentiment markers. The model therefore recognizes explicit polarization reliably but misses implicit polarization.

Interestingly, the number of false positives remains low. This indicates that the model does not simply equate strong sentiment with polarization.

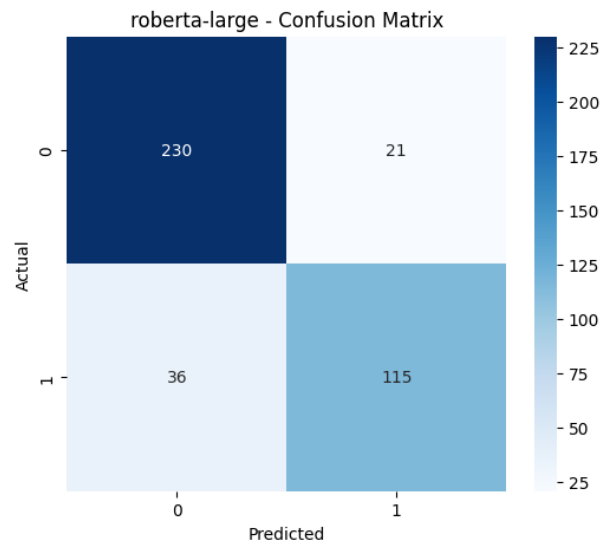


Figure 9: Confusion Matrix of RoBERTa-Large for POLAR subtask-1

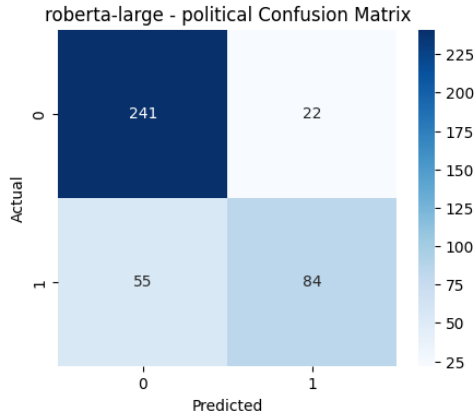
Instead, it has learned a relatively specific notion of polarized discourse. From a moderation perspective, this is a desirable bias: false positives would incorrectly flag normal discussions, whereas false negatives mainly reduce recall.

Another noticeable pattern is the imbalance effect. Because the dataset contains more non-polarized than polarized samples, the classifier learns a slightly higher decision threshold in practice, even though the theoretical threshold is 0.5. This explains why polarized content that is subtle or context-dependent is frequently classified as neutral.

6.3 Multi-label Task: Confusion Matrix and Error Patterns

The per-class confusion matrices for Subtask 2 (Figure 10) reveal a much more complex failure mode. The dominant political category is predicted with relatively high recall, while minority categories—especially gender/sexual orientation and religious polarization—are frequently missed. Notably, the dataset is extremely imbalanced: for instance, there are only 7 gender/sexual orientation examples and a similarly small number of religious polarization examples. This scarcity amplifies the difficulty of learning these categories.

The main issue is not incorrect classification into the wrong category but rather *label omission*. The model often predicts only the political label even when multiple labels are present. This directly reflects the dataset’s label co-occurrence: many



(a) Roberta-Political polarization confusion matrix

figure/roberta_gender_CM.png

(b) Roberta-Gender/Sexual orientation confusion matrix

Figure 10: RoBERTa-Large confusion matrices for two polarization categories.

messages simultaneously express political and social bias. Because political polarization appears most frequently, the model learns it as the safest prediction.

This creates a systematic effect. When multiple labels should be active, the model tends to predict only the most probable one. From the perspective of binary classification per label, this appears as false negatives for minority classes. Consequently, macro-F1 drops significantly even though the model often partially understands the message.

The gender/sexual category shows the lowest recall. This is not a random failure but a linguistic and data issue. Gender-related polarization often uses indirect references, humor, or cultural stereotypes rather than explicit offensive terms. Combined with the extremely limited number of examples, the model struggles to learn meaningful patterns. Transformers trained on general corpora are better at recognizing political discourse patterns than socially coded language, which requires more cultural context.

Racial and religious categories show moderate precision but low recall. The model detects clear discriminatory statements but struggles with ambiguous or contextual ones. As with gender/sexual orientation, the scarcity of minority examples—most labels are 0—further hinders learning, suggesting the classifier relies heavily on lexical cues rather than deeper pragmatic reasoning.

6.4 Effect of Class Imbalance and Focal Loss

Class weighting and focal loss noticeably improve minority category detection but do not solve the problem completely. The loss modification encourages the model to pay attention to rare examples; however, the limitation is not purely statistical. The difficulty lies in semantic overlap between labels. For instance, a message criticizing immigration policy may simultaneously contain political and racial polarization. The model must learn a multi-dimensional interpretation of meaning, which is inherently harder than standard classification.

Because evaluation uses macro-F1, errors on rare categories significantly reduce the score. The large gap between binary performance and multi-label performance therefore does not indicate model failure but reflects the intrinsic complexity of categorizing social meaning.

7 Conclusion and Future Work

Our experiments indicate that detecting *whether* a message is polarized is largely a stance detection task, whereas identifying *what type* of polarization it represents requires deeper social reasoning. Transformer-based models excel at capturing explicit stance signals, such as agreement, disagreement, or hostility, but struggle to infer the underlying social context or categorize polarization according to sociological dimensions.

Analysis of the confusion matrices reveals that while the models can reliably detect overt hostility and opposition, they frequently fail when multiple social identities or nuanced relational dynamics are involved, highlighting a gap in their social reasoning capabilities. This issue is further exacerbated by the severe imbalance in the dataset: minority categories such as gender/sexual orientation or religious polarization are extremely rare (most labels are 0), which leads the model to favor the dominant political category. As a result, when multiple labels should be active, the model tends to predict only the most frequent one, producing systematic false negatives and lowering macro-F1 scores despite partially correct understanding of the message.

These findings suggest that effective polarization detection cannot rely solely on linguistic patterns; it also requires models to encode social context, relationships, and the interplay between identities. Future work could explore integrating structured social knowledge, such as identity hierarchies, social role embeddings, or knowledge graphs, into

model architectures to enhance categorization performance. Additionally, multi-task learning frameworks could be employed to jointly learn stance detection and social categorization, potentially enabling the model to leverage shared representations. Contrastive pretraining techniques, where the model learns to distinguish between different types of polarized content, may further improve its ability to capture subtle social signals. Finally, incorporating human-in-the-loop approaches or explainability mechanisms could help identify and address systematic biases, ultimately leading to more robust and socially aware polarization detection systems.

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8 Appendix

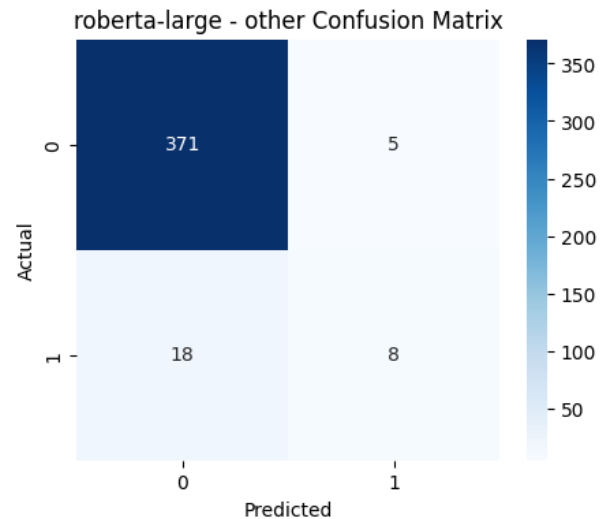


Figure 11: Confusion Matrix of RoBERTa-Large for Others

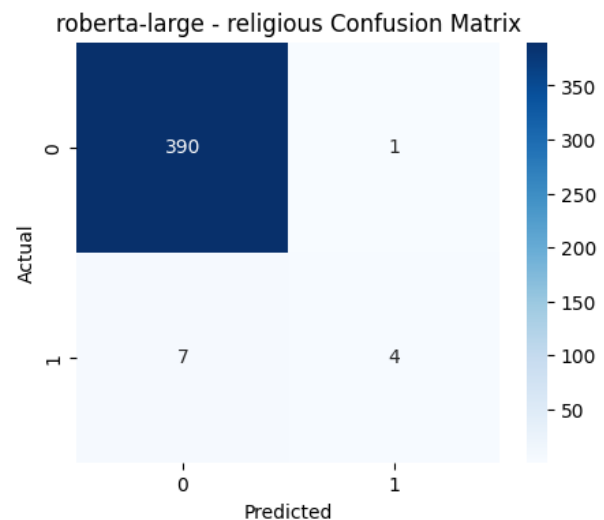


Figure 12: Confusion Matrix of RoBERTa-Large for religion